PSYC4411

DISCOVERING STRUCTURE

CLUSTERING & DIMENSIONALITY REDUCTION

Week 7 · Semester 1, 2026

TODAY'S PLAN

1. The shift to unsupervised learning — no right answers
2. Dimensionality reduction — PCA and the scree plot
3. UMAP & t-SNE — vivid but dangerous visualisations
4. Clustering — k-means, hierarchical, DBSCAN
5. Stability — the most important question in clustering
6. The psychology of "types" — categories vs. dimensions
7. Getting ready for Week 8

THE SHIFT TO UNSUPERVISED LEARNING

When there's no "right answer"

SUPERVISED VS. UNSUPERVISED

SUPERVISED

Weeks 2–6

You have a **target variable**

"Predict depression score"

"Classify elevated vs. minimal"

Clear right answers → clear metrics

→

UNSUPERVISED

This week

No target variable

"Are there subgroups?"

"What are the latent dimensions?"

No right answers → stability matters

TWO MAIN TASKS

DIMENSIONALITY REDUCTION

Simplify complex data into fewer dimensions while keeping the important patterns.

42 questionnaire items → 3 components

CLUSTERING

Discover groups of similar observations without being told the groups in advance.

“Are there depression subtypes?”

If you’ve run a **factor analysis** in statistics, you’ve already done a version of unsupervised ML.

THE DANGER OF EXPLORATION

Clustering algorithms will ALWAYS find groups — even in random noise.

- Humans are pattern-seeking creatures
- Ask for 3 clusters → you'll get 3 clusters, guaranteed
- The question isn't "did I find clusters?"
- It's "are these clusters **real, stable, and meaningful**?"

This week's core skill: distinguishing **discovery** from **fabrication**.

SAME DATA, DIFFERENT QUESTION

WEEK 4 (SUPERVISED)

DASS-42 dataset

"Can personality **predict** depression scores?"

Target: DASS_Depression

Metrics: MAE, R^2

WEEK 8 (UNSUPERVISED)

Same dataset

"Are there **subgroups** of distress?"

No target variable

Metrics: silhouette, stability

DIMENSIONALITY REDUCTION

Seeing the big picture

THE CURSE OF DIMENSIONALITY

- The DASS-42 has **42 items** — each person lives in 42-dimensional space
- You can't visualise 42 dimensions
- Models need exponentially more data as dimensions increase
- Many dimensions contain mostly **noise**, not signal

Dimensionality reduction finds a simpler view that preserves the important patterns — like summarising a long conversation.

PRINCIPAL COMPONENT ANALYSIS (PCA)

Week 7

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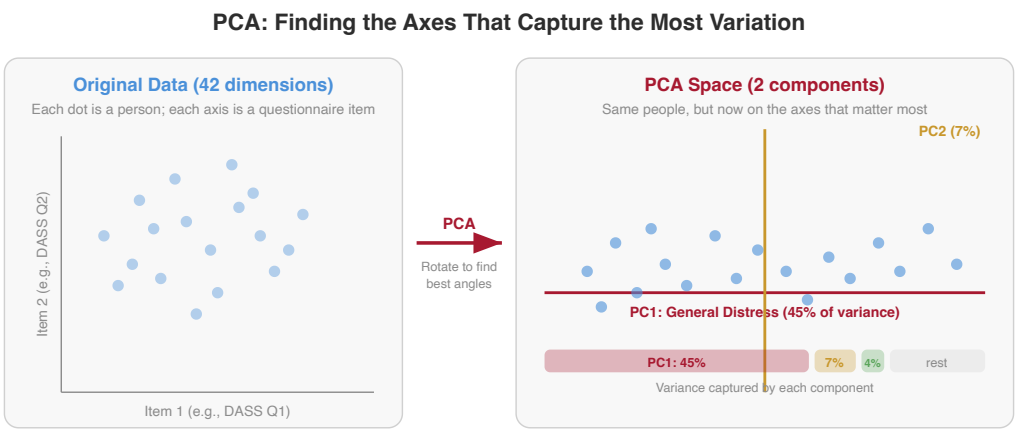
- PCA finds new axes — **principal components** — that capture the most variation
- PC1 captures the single direction of **greatest variation**
- PC2 captures the next most (perpendicular to PC1)
- And so on...

Analogy: Photographing a coffee mug from every angle. Most photos look similar. PCA finds the *few angles* that show the most different views.

PCA VISUALLY



42 questionnaire items → 3 components capture ~56% of all variation

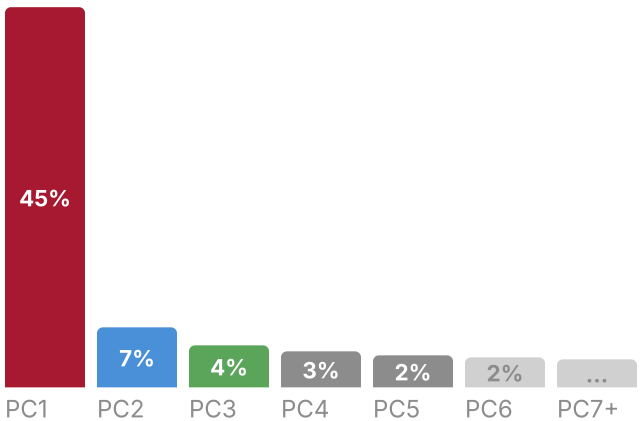
PCA doesn't discard data — it finds a simpler view that preserves the most important patterns

Like photographing a sculpture from its most informative angles

42 questionnaire items → 3 components that capture 56% of all variation. PCA doesn't discard data — it finds a **simpler view** that preserves the most important patterns.

THE SCREE PLOT

How much variance does each component capture?



DASS-42: 3 components capture **~55%** of all variation. 42 dimensions → 3.

WHAT DOES “45% OF VARIANCE” ACTUALLY MEAN?

Without PCA

To describe someone’s distress pattern, you’d need all **42 item scores**. That’s a lot of numbers.

With PC1 alone

One single number — their score on PC1 — tells you nearly **half the story**. It’s like a master volume dial for distress.

“Variance explained” = how much of the **total spread** in scores this dimension captures.

45% from one component is **unusually high** — it means distress is more “one big thing” than “42 separate things.”

If everyone who scores high on one distress symptom also scores high on the others, a single “general distress” factor will capture most of the variation. That’s exactly what PC1 is.

LOADINGS: WHAT DOES EACH COMPONENT MEAN?

PC1: General Distress

45% of variance

High loadings on **all** DASS items

People who score high on one symptom tend to score high on everything

PC2: Depression vs. Anxiety

7% of variance

Depression items load **positive**, Anxiety items load **negative**

What distinguishes depression from anxiety

PC3: Stress vs. Anxiety

4% of variance

Stress items load **positive**, Anxiety items load **negative**

What distinguishes stress from anxiety

The DASS three-factor structure emerges — but PC1 (general distress) dominates everything else.

THINK ABOUT IT

A researcher runs PCA on 42 DASS items and finds 3 components. They call these "Depression," "Anxiety," and "Stress" and say they've **"discovered"** the three-factor structure.

But those are the three subscales the questionnaire was *designed* to measure.

Have they discovered structure — or **rediscovered** something built into the questionnaire from the start?

UMAP & T-SNE

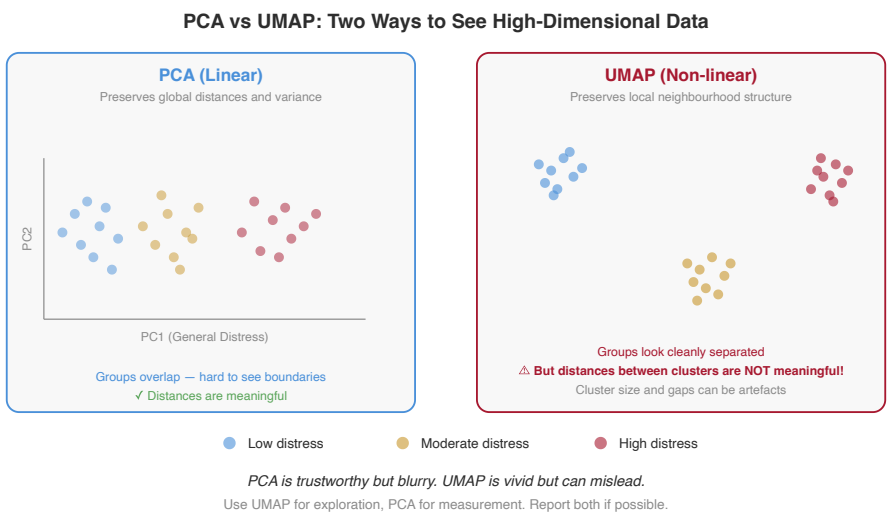
Non-linear visualisation

PCA IS LINEAR

- PCA finds straight-line relationships (linear combinations)
- But what if groups are separated by **curves**, not lines?
- What if the structure is more complex than linear axes can capture?

UMAP (Uniform Manifold Approximation and Projection) finds **non-linear** structure and creates 2D maps where similar points cluster together.

PCA VS. UMAP: SAME DATA, VERY DIFFERENT PICTURES



PCA is **trustworthy but blurry** — distances are real. UMAP is **vivid but can mislead** — cluster gaps and sizes may be artefacts. Use UMAP for exploration, PCA for measurement.

PCA VS. UMAP: QUICK COMPARISON

PCA (LINEAR)

Groups overlap, boundaries blurry

- Distances are meaningful
- Reproducible (no randomness)
- Variance explained is quantifiable

Best for: measurement, inference, reporting

UMAP (NON-LINEAR)

Groups look clean and well-separated

- Distances between clusters are NOT meaningful
- Cluster sizes and shapes can be artefacts
- Results change with different settings

Best for: exploration, hypothesis generation

UMAP is a **sketch**, not a photograph. A map for exploration, not a measurement tool for inference.

UMAP: HANDLE WITH CARE

Three things UMAP can distort:

1. **Distances between clusters** — two groups far apart on the UMAP plot might actually be similar
2. **Cluster sizes** — a large blob might represent fewer people than a small blob
3. **Cluster shapes** — tight circles might actually be elongated in the original space

Only **local structure** is preserved: nearby points stay nearby. Everything else is up for distortion.

UMAP is a **sketch**, not a photograph. Use it for exploration, not confirmation.

CLUSTERING

Finding groups in data

WHAT MAKES A “GOOD” CLUSTER?

Cohesion

Points **within** a cluster are **similar** to each other

Separation

Points in **different** clusters are **dissimilar**

Remember: clustering algorithms will **always** find groups, even in random data. The question is whether the groups are **real**.

THREE ALGORITHM FAMILIES

k-Means

Centroid-based

Define clusters around central points. Fast and intuitive.

You choose k in advance

Hierarchical

Agglomerative

Build a tree of nested groups. See the full hierarchy.

No need to pre-specify k

DBSCAN

Density-based

Find dense regions. Handles odd shapes and noise.

Identifies “noise” points

K-MEANS: HOW IT WORKS

1

Place Centroids

Randomly place k central points in the data space

2

Assign Points

Each data point joins its nearest centroid

3

Move Centroids

Each centroid moves to the mean of its assigned points

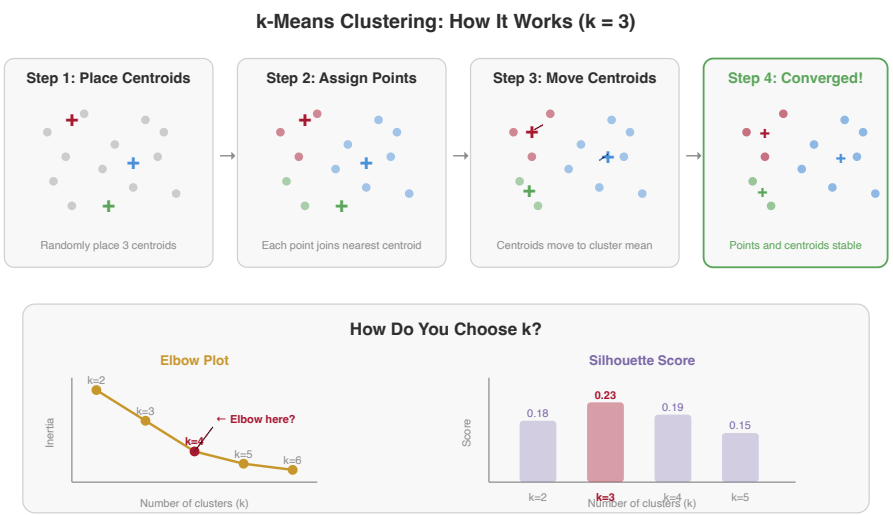
4

Repeat

Steps 2–3 until nothing changes (convergence)

Fast, intuitive, and usually the first method to try. But assumes roughly spherical, equal-sized clusters.

K-MEANS: WATCHING IT WORK



The "right" number of clusters is a judgement call, not a mathematical fact — these tools guide the decision

The **data points stay put** — only the centroids move. The algorithm converges when nobody changes cluster. Different random starting positions can give different results, which is why **stability checks** matter.

CHOOSING K

Elbow Plot

Plot within-cluster distance vs. k

Look for the **“elbow”** — where adding more clusters stops helping much

Silhouette Score

How well does each point fit its cluster vs. the nearest other cluster?

Range: **−1** (bad) to **+1** (perfect)

Neither tool gives a definitive answer. The “right” number of clusters is a **judgement call** — guided by diagnostics and by whether the clusters make **psychological sense**.

EXAMPLE: DASS-42 CLUSTERING

SILHOUETTE SCORES

k	Silhouette
2	**0.240**
3	0.150
4	0.145
5	0.132

k = 2 wins — but even 0.24 is modest

TWO CLUSTERS

Cluster 0 (N = 16,908): Low distress

Depression M = 11.8, Anxiety M = 8.6

Cluster 1 (N = 17,668): High distress

Depression M = 29.7, Anxiety M = 22.8

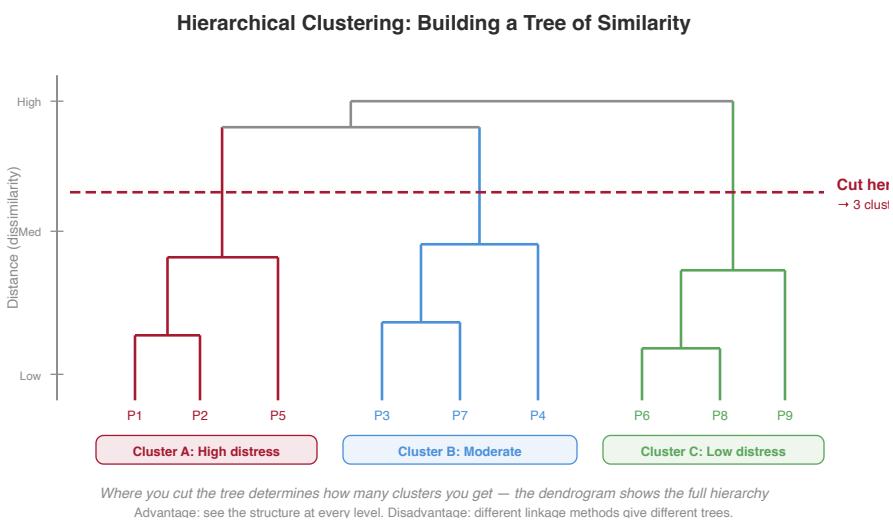
HIERARCHICAL CLUSTERING

- Start with every point as its own cluster
- Merge the two most similar clusters at each step
- Continue until everything is in one cluster
- Result: a **dendrogram** (tree) showing the full merge history

Advantage: See the full hierarchy. Cut at any height to get any number of clusters.

Weakness: Different linkage methods (single, complete, Ward's) give different trees — another source of instability.

READING A DENDROGRAM



- **Height** = how different the merged clusters are
- **Cut horizontally** to choose the number of clusters
- Cut high → few broad groups. Cut low → many small groups.
- You can see the **full hierarchy** — no need to commit to one k

Advantage: See nested structure — maybe 2 broad groups, each containing 2 subgroups.

Weakness: Different linkage methods give different trees — another source of instability.

DBSCAN: DENSITY-BASED CLUSTERING

- Clusters = regions of **high density** separated by low density
- Two parameters: **epsilon** (neighbourhood radius) and **min_samples** (minimum cluster size)
- Points that don't belong to any cluster are labelled **"noise"**

Strengths: No need to specify k. Finds odd shapes. Identifies outliers.

Weaknesses: Sensitive to parameters. Struggles with varying density.

STABILITY

Can you trust your clusters?

THE MOST IMPORTANT QUESTION

If you run the analysis again, do you get the same clusters?

- Different **random seed** → same clusters?
- Random **80% subsample** (bootstrap) → same memberships?
- Different **algorithm** entirely → same groups?

A cluster solution that changes every time you look at it isn't a discovery — it's **noise**.

DASS-42 STABILITY CHECK

k = 2 solution across 5 random seeds:

Seed	Points Changed	% Changed
42 (reference)	—	—
123	116	0.34%
456	65	0.19%
789	54	0.16%
2026	2	0.01%

This solution is **very stable** — fewer than 0.35% of participants change cluster. But k = 2 is also simple (low vs. high distress). More nuanced solutions (k = 4, 5) would be less stable.

SILHOUETTE SCORES

- For each point: how much closer is it to its **own cluster** than to the **nearest other cluster**?
- Range: −1 (badly placed) to +1 (perfectly placed)
- Average across all points = overall cluster quality

Score Range	Interpretation
0.70 – 1.00	Strong structure (rare in psychology)
0.50 – 0.70	Reasonable structure
0.25 – 0.50	Weak structure (common in psychology)
< 0.25	No meaningful structure

Our k = 2 DASS solution: silhouette = **0.24** — right at the boundary. The clusters exist, but they're not strongly separated.

THINK ABOUT IT

You cluster participants into 3 groups based on their depression and anxiety patterns. At a conference, someone asks: **“Did you check whether those clusters are stable?”**

You rerun with a different random seed and get **4 groups** with different boundaries.

What should you do next? What does this tell you about the “reality” of the subgroups?

THE PSYCHOLOGY OF “TYPES”

Categories vs. dimensions

PSYCHOLOGY LOVES CATEGORIES

- **Personality types:** introvert/extrovert, Type A/B
- **Diagnostic categories:** MDD vs. GAD vs. PTSD
- **Learning styles:** visual/auditory/kinesthetic

Some are useful: Diagnoses guide treatment. Categories simplify communication.

Some lack evidence: Learning styles have been extensively debunked.

But are these categories **real** — or convenient fictions?

THE TAXOMETRICS DEBATE

CATEGORICAL

You **have** depression or you **don't**

Discrete types, clear boundaries

vs.

DIMENSIONAL

Everyone sits **somewhere** on a severity continuum

No natural boundaries

Haslam et al. (2020): Meta-analysis of 317 taxometric studies. Dimensional models outnumber categorical 5:1 across psychology. Most psychopathology is **dimensional**.

THE DANGER OF REIFICATION

Reification: naming a cluster makes it *feel* real, even if it barely survives a random seed change.

- You find 3 clusters and call them “The Anxious Achiever,” “The Resilient Introvert,” “The Stressed Coper”
- These labels **create the illusion of categories** where continuous variation might be a better description
- If the clusters change with a different random seed, the names were **premature**

Names are powerful. Use them **after** stability checks, not before.

BORSBOOM'S NETWORK APPROACH

A third perspective: symptoms **cause each other** rather than being caused by a latent disorder.

Poor sleep → **Low energy** → **Withdrawal** → **Loneliness** → **Sad mood**

> Depression isn't a hidden disease entity — it's a **self-reinforcing network** of mutually activating symptoms. This challenges both the categorical and dimensional frameworks.

THINK ABOUT IT

Mental health researchers have debated for decades whether conditions like depression are discrete **“types”** or points on a **continuum**.

Could clustering analysis settle this debate?

What would you need to see in the data to be convinced that **discrete types genuinely exist**?

COMMON MISCONCEPTIONS

“PCA discovers hidden factors.”

PCA finds linear combinations that maximise variance. Whether these correspond to meaningful psychological constructs depends on **interpretation**, not mathematics.

“More clusters = better model.”

Adding clusters *always* improves fit on the training data. The question is whether additional clusters are **stable and meaningful**.

“UMAP shows the true structure.”

UMAP is optimised for visualisation. It can distort distances, densities, and cluster boundaries. Treat it as a **sketch**, not a photograph.

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GETTING READY FOR WEEK 8

PCA, UMAP, and clustering on real data

WEEK 8: CHALLENGE LAB (SELF-PACED)

📅 **Monday 27 April is a public holiday** — no class next week. Complete the Week 8 lab in your **assigned groups** in your own time.

- Same DASS-42 dataset from Week 4 — different question
- **PCA** on 42 distress items — how many components?
- **UMAP** visualisation — colour by depression severity
- **k-Means clustering** on personality + distress profiles
- **Stability checks** — different seeds, silhouette scores
- **Write a methods paragraph** with your AI assistant

You may still have the data from Week 4. If not: `cd weeks/week-08-lab/data && python download_data.py`

NEW LLM SKILL: DOCUMENTATION

In research, your analysis is only as good as your ability to **explain it**.

Weak prompt

"Explain my code."

Strong prompt

"I ran PCA on 42 DASS items from 34,576 participants. 3-component solution explaining 55% of variance. Write a methods paragraph for a psychology journal. APA style. Include sample size, measures, software."

> The AI's draft is a **starting point**. You must verify every number and method name matches what you actually did.

QUESTIONS?

Week 8: PCA, UMAP & Clustering Lab